2. Create “Hello World” application. That will display “Hello World” in the middle of the screen using TextView Widget in the red color.

AndroidMenifest.xml

```
<activity android:name=".HelloWorldActivity" android:exported="true" >
    <intent-filter >
        <action android:name="android.intent.action.MAIN" />
        <category android:name="android.intent.category.LAUNCHER "
tools:node="merge"/>
    </intent-filter>
</activity>
```

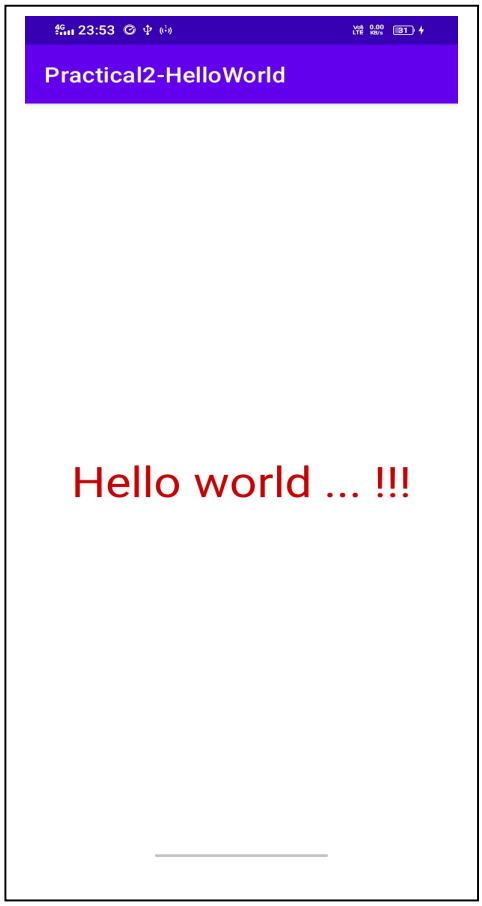
HelloWorldActivity.java

```
public class HelloWorldActivity extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.hello_world_layout);
    }
}
```

hello_world_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android=
"http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    >

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:gravity="center"
        android:text="Hello world ... !!!"
        android:textColor
="@android:color/holo_red_dark"
        android:textSize="40sp" />

</LinearLayout
```

OUTPUT :



3. Create Registration page to demonstration of Basic widgets available in android.

AndroidMenifest.xml

```
<activity android:name=".RegistrationPageActivity" android:exported="true">
    <intent-filter >
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>
```

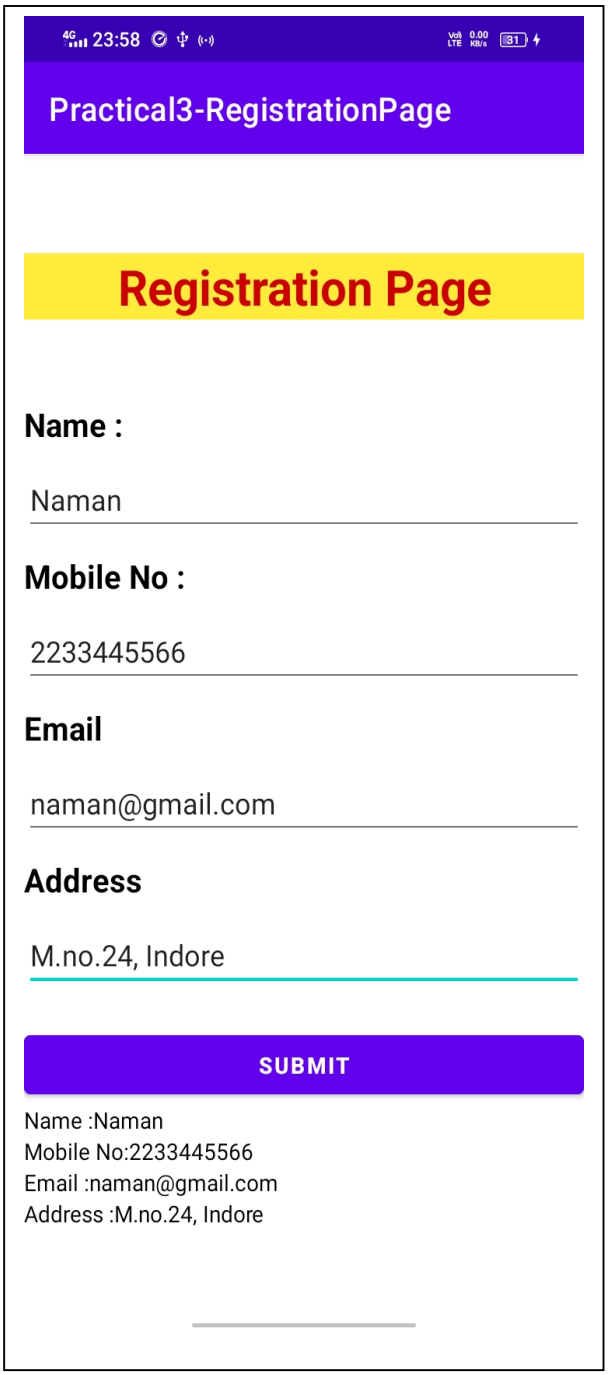
RegistrationPageActivity.java

```
    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.registration_page_layout);

        Button btnSubmit = findViewById(R.id.btnSubmit);

        btnSubmit.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {

                EditText txtName = findViewById(R.id.txtName);
                EditText txtMobileNo  = findViewById(R.id.txtMobileNo);
                EditText txtEmail = findViewById(R.id.txtEmail);
                EditText txtAddress = findViewById(R.id.txtAddress);

                TextView lblName = findViewById(R.id.lblName);
                TextView lblMobileNo = findViewById(R.id.lblMobileNo);
                TextView lblEmail = findViewById(R.id.lblEmail);
                TextView lblAddress = findViewById(R.id.lblAddress);

                lblName.setText(lblName.getText().toString() +
txtName.getText().toString());
                lblMobileNo.setText(lblMobileNo.getText().toString() +
txtMobileNo.getText().toString());
                lblEmail.setText(lblEmail.getText().toString() +
txtEmail.getText().toString());
                lblAddress.setText(lblAddress.getText().toString() +
txtAddress.getText().toString());

            }
        });
    }
```

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android
="http://schemas.android.com/apk/res/androi
d"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <TextView
        android:layout_width="match_parent"

android:layout_height="wrap_content"
        android:background="#FFEB3B"
        android:gravity="center"
        android:text="Registration Page"
        android:textColor
="@android:color/holo_red_dark"
        android:textSize="30sp"
        android:textStyle="bold"
        android:layout_marginBottom="40dp"
        />
    <TextView
        android:paddingVertical="10dp"
        android:layout_width="match_parent"

android:layout_height="wrap_content"
        android:text="Name :"
        android:textColor="@color/black"
        android:textSize="20dp"
        android:textStyle="bold"
        />

    <EditText
        android:id="@+id/txtName"
        android:layout_width="match_parent"

android:layout_height="wrap_content"
        android:hint="Enter full name " />

    <TextView
        android:paddingVertical="10dp"
        android:layout_width="match_parent"

android:layout_height="wrap_content"
        android:text="Mobile No :"
        android:textColor="@color/black"
        android:textSize="20dp"
        android:textStyle="bold"
        />

    <EditText
        android:id="@+id/txtMobileNo"
        android:layout_width="match_parent"

android:layout_height="wrap_content"
        android:hint="Enter Mobile"/>


    <TextView
        android:paddingVertical="10dp"
        android:layout_width="match_parent"

android:layout_height="wrap_content"
```

```xml
<EditText
        android:id="@+id/txtEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter email"/>

    <TextView
        android:paddingVertical="10dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Address"
        android:textColor="@color/black"
        android:textSize="20dp"
        android:textStyle="bold"
        />

    <EditText
        android:id="@+id/txtAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Address"/>

    <Button
        android:id="@+id/btnSubmit"
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Submit"
        android:textStyle="bold"
        />

    <TextView
        android:id="@+id/lblName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Name :"
        android:textColor="@color/black"
        />

    <TextView
        android:id="@+id/lblMobileNo"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Mobile No:"
        android:textColor="@color/black"
        />

    <TextView
        android:id="@+id/lblEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Email :"
        android:textColor="@color/black"
        />

    <TextView
        android:id="@+id/lblAddress"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Address :"
        android:textColor="@color/black"
        />

</LinearLayout>
```

OUTPUT


23:58
Practical3-RegistrationPage
Registration Page
Name :
Naman
Mobile No :
2233445566
Email
naman@gmail.com
Address
M.no.24, Indore
SUBMIT
Name :Naman
Mobile No:2233445566
Email :naman@gmail.com
Address :M.no.24, Indore

4. Create sample application with login module.(Check username and password) On successful login, Change TextView "Login Successful". And on failing login, alert user using Toast "Login fail".

AndroidMenifest.xml

```
<activity android:name=".LoginActivity" android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>
```

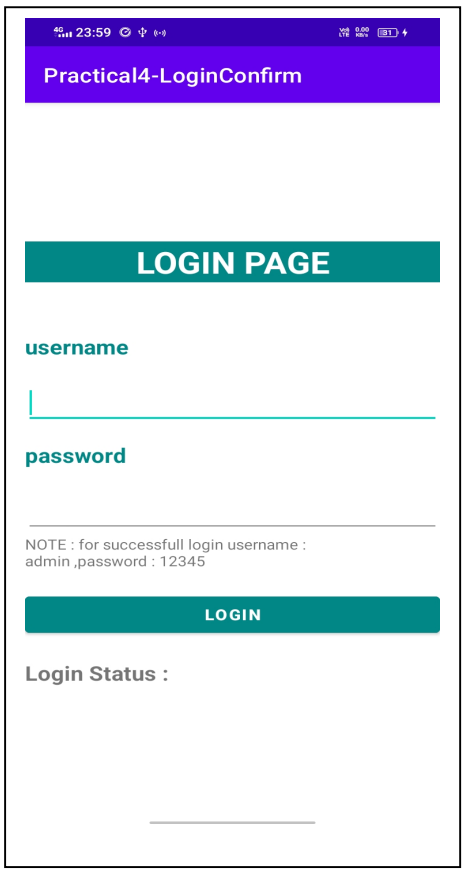
LoginActivity.java

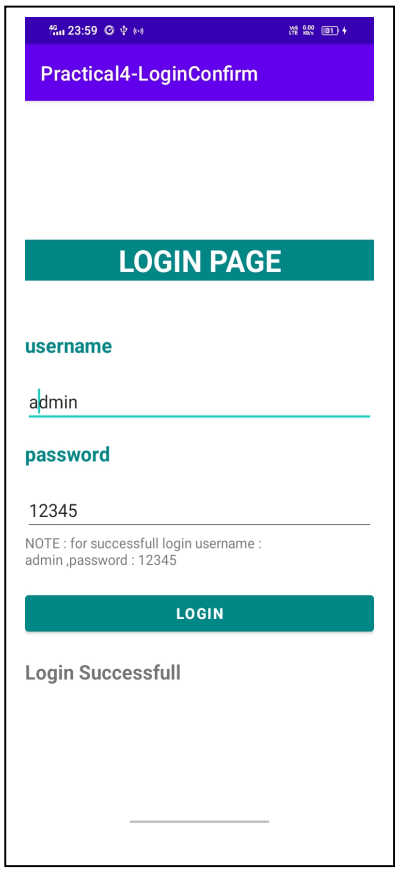
```
@Override
protected void onCreate(@Nullable Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.login_layout);

    Button button = findViewById(R.id.btnLogin) ;

    button.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View v) {

            EditText txtUsername = findViewById(R.id.txtUsername);
            EditText txtPassword = findViewById(R.id.txtPassword);
            TextView txtStatus = findViewById(R.id.lblLoginStatus);
            boolean flag = false ;

            if(txtUsername.getText().toString().equals("admin") &&
txtPassword.getText().toString().equals("12345")) {
                txtStatus.setText("Login Successfull");
                Toast.makeText(getApplicationContext(),"Login
Successfull",Toast.LENGTH_SHORT).show();
            }else {
                txtStatus.setText("Login Fail");
                Toast.makeText(getApplicationContext(), "Login Fail",
Toast.LENGTH_SHORT).show();
            }
        }
    });
}
```

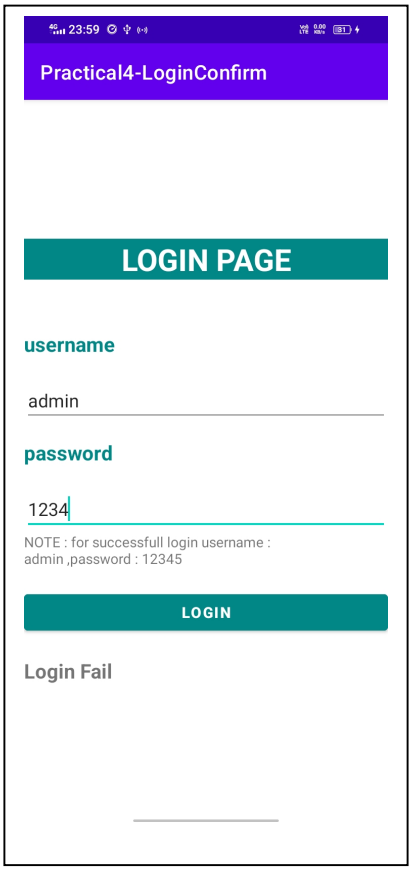
login_layout.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android=
"http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:backgroundTint="@color/teal_700"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginBottom="50dp"
        android:background="@color/teal_700"
        android:backgroundTint="@color/teal_700"
        android:gravity="center"
        android:text="LOGIN PAGE"
        android:textColor="@color/white"
        android:textSize="30dp"
        android:textStyle="bold" />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="username"
        android:textColor="@color/design_defaul
       t_color_secondary_variant"
        android:textSize="20dp"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/txtUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp" />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="15dp"
        android:text="password"
        android:textColor="@color/design_
       default_color_secondary_variant"
        android:textSize="20dp"
        android:textStyle="bold" />

    <EditText
        android:id="@+id/txtPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_marginTop="20dp" />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text=
       "NOTE : for successfull login
       username : admin ,password : 12345"
        />

<Button
        android:id="@+id/btnLogin"

android:layout_width="match_parent"

android:layout_height="wrap_content"

android:layout_marginTop="20dp"

android:backgroundTint="@color/teal_70
0"
        android:text="LOGIN"
        android:textSize="15dp"
        android:textStyle="bold" />


<TextView

android:layout_marginTop="20dp"

android:id="@+id/lblLoginStatus"

android:layout_width="match_parent"

android:layout_height="wrap_content"
        android:textSize="20dp"
        android:textStyle="bold"
        android:text="Login Status :"
        />

</LinearLayout>
```

OUTPUT :


23:59
Practical4-LoginConfirm
LOGIN PAGE
username
password
NOTE : for successfull login username :
admin ,password : 12345
LOGIN
Login Status :


23:59
Practical4-LoginConfirm
LOGIN PAGE
username
admin
password
12345
NOTE : for successfull login username :
admin ,password : 12345
LOGIN
Login Successfull


23:59
Practical4-LoginConfirm
LOGIN PAGE
username
admin
password
1234
NOTE : for successfull login username :
admin ,password : 12345
LOGIN
Login Fail

5. Create an application for demonstration of Scroll view in android.

AndroidMenifest.xml

```
<activity android:name=".MainActivity" android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>

</activity>

<activity android:name=".VeriticalScrlView" android:exported="true"/>
<activity android:name=".HorizontalScrlViewActivity" android:exported="true"/>
```

VeriticalScrlView.java

```
public class VeriticalScrlView extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.vertical_scrlview_layout);
    }
}
```

vertical_scrlview_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android=
"http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        >

        <TextView
            android:layout_width="match_parent"

android:layout_height="wrap_content"

android:background="@color/teal_700"
            android:gravity="center"
            android:text="Vertical ScrollView"
            android:textColor=
"@color/material_dynamic_neutral_variant100"
            android:textSize="30dp"
            android:textStyle="bold" />

        <TextView
            android:layout_width="match_parent"
            android:layout_height="335dp"
            android:layout_marginBottom="20dp"
            android:text="Hello World ...!!!"
            android:textSize="50dp"
            android:textStyle="bold" />

<TextView
android:layout_marginTop="70dp"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textStyle="bold"
android:textSize="20dp"
android:text="Sample of
VerticalScrollView"
android:layout_marginBottom="70dp" />

<TextView
android:layout_marginTop="70dp"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textStyle="bold"
android:textSize="50dp"
android:text="Scroll Page"
android:layout_marginBottom="70dp" />

<TextView
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="70dp"
android:layout_marginBottom="70dp"
android:text="Hiii ... Enjoy android "
android:textSize="50dp"
android:textStyle="bold" />

</LinearLayout>

</ScrollView>
```

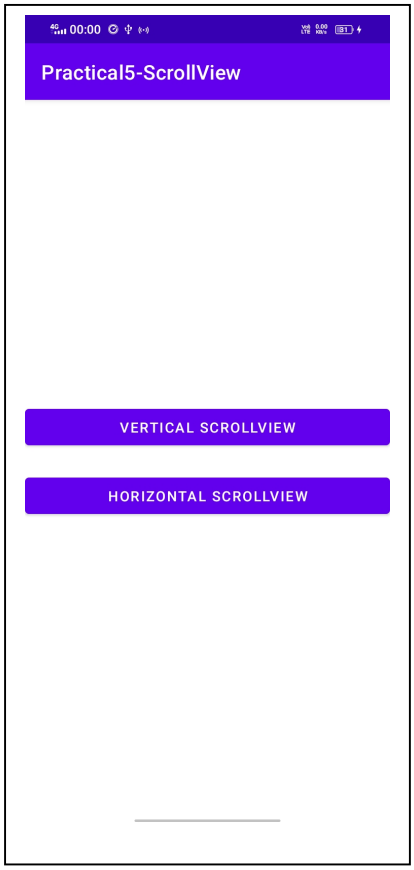
HorizontalScrlViewActivity.java

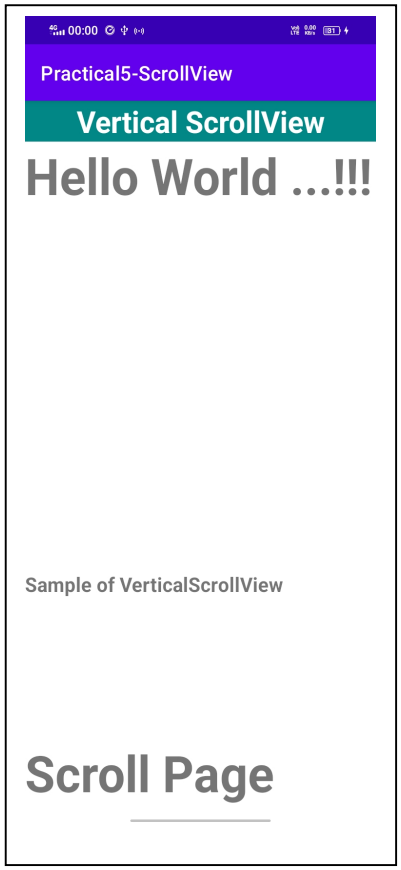
```java
public class HorizontalScrlViewActivity extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.horizontal_scrlview_layout);
    }
}
```

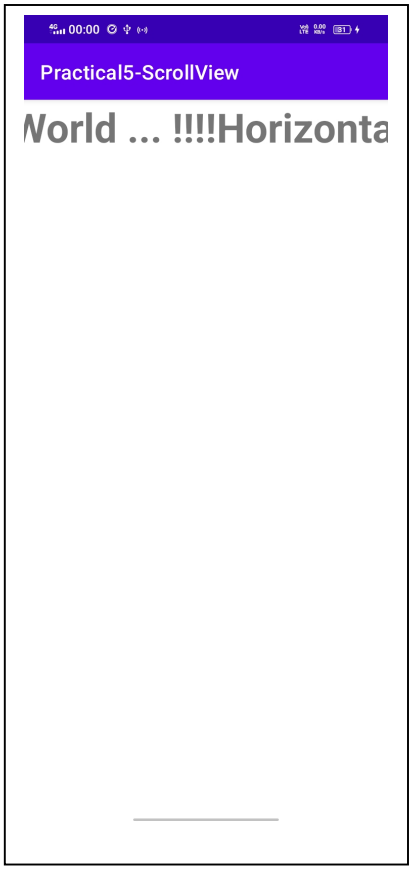
horizontal_scrlview_layout.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<HorizontalScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Hello World ... !!!!"
            android:textSize="40dp"
            android:textStyle="bold"
            />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Horizontal ScrollView
"
            android:textSize="40dp"
            android:textStyle="bold"
            />
    </LinearLayout>
</HorizontalScrollView>
```

OUTPUT :


00:00
Practical5-ScrollView
VERTICAL SCROLLVIEW
HORIZONTAL SCROLLVIEW


00:00
Practical5-ScrollView
Vertical ScrollView
Hello World ...!!!
Sample of VerticalScrollView
Scroll Page


00:00
Practical5-ScrollView
Vorld ... !!!!Horizonta

6. Create login application where you will have to validate username and passwords till the username and password is not validated, login button should remain disabled.

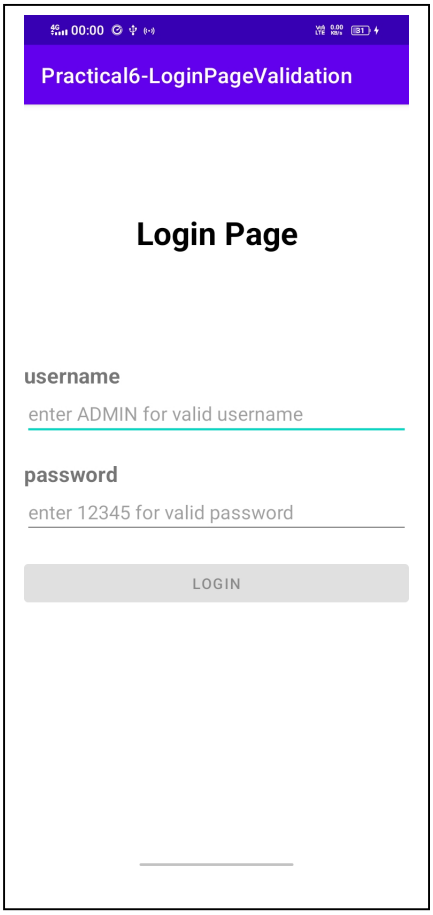
AndroidMenifest.xml

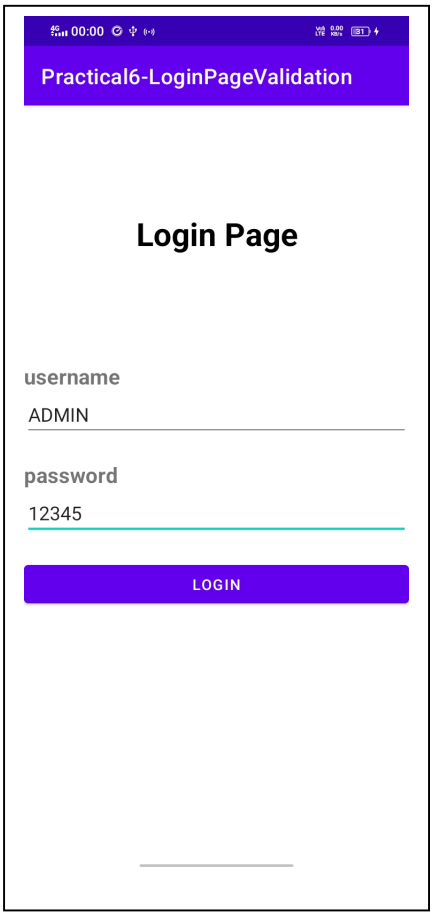
```
<activity android:name=".LoginPageActivity" android:exported="true">
    <intent-filter>
        <category android:name="android.intent.category.LAUNCHER"/>
        <action android:name="android.intent.action.MAIN"/>
    </intent-filter>
</activity>
```

LoginPageActivity.java

```
public class LoginPageActivity extends AppCompatActivity {
    private EditText txtUsername ;
    private EditText txtPassword;
    private Button btnLogin;

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.login_page_layout);

        txtUsername = findViewById(R.id.txtUsername);
        txtPassword = findViewById(R.id.txtPassword);
        btnLogin = findViewById(R.id.btnLogin);

        txtUsername.addTextChangedListener(new TextWatcher() {
            @Override
            public void beforeTextChanged(CharSequence s, int start, int count, int after) {}

            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
                enableButton();
            }

            @Override
            public void afterTextChanged(Editable s) {}
        });

        txtPassword.addTextChangedListener(new TextWatcher() {
            @Override
            public void beforeTextChanged(CharSequence s, int start, int count, int after) {}

            @Override
            public void onTextChanged(CharSequence s, int start, int before, int count) {
                enableButton();
            }

            @Override
            public void afterTextChanged(Editable s) {}
        });
    }
        private void enableButton(){

        String username = txtUsername.getText().toString();
        String password = txtPassword.getText().toString();

        if(username.equals("ADMIN") && password.equals("12345")){
            btnLogin.setEnabled(true);
        }else{
            btnLogin.setEnabled(false);
        }
    }
```

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android=
"http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
     >

    <TextView
        android:layout_marginTop="100dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login Page"
        android:textSize="30dp"
        android:textStyle="bold"
        android:textColor="@color/black"
        android:gravity="center"
        android:layout_marginBottom="100dp"
        />

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="username"
        android:textSize="20dp"
        android:textStyle="bold"
        />

    <EditText
        android:id="@+id/txtUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="enter ADMIN for valid
username"
        />

    <TextView
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="password"
        android:textSize="20dp"
        android:textStyle="bold"

        />

    <EditText
        android:id="@+id/txtPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="enter 12345 for valid
password"
        />

    <Button
        android:id="@+id/btnLogin"
        android:layout_marginTop="20dp"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="login"
        android:enabled="false"

        />

</LinearLayout>
```





7. Create an application for calculator.

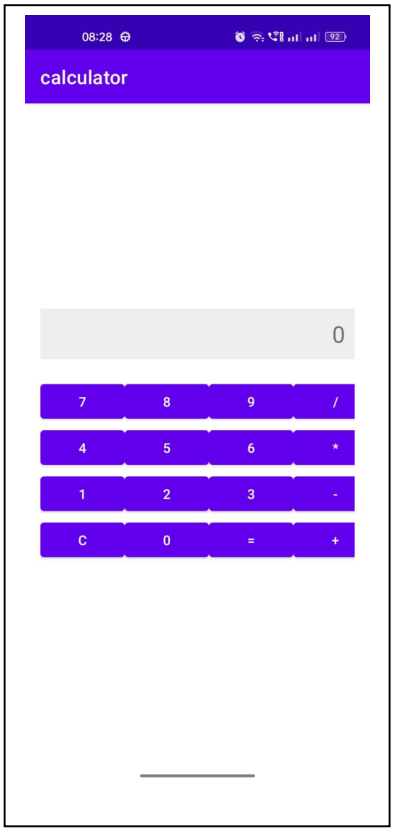
AndroidMenifest.xml

```
<activity android:name=".MainActivity"
        android:exported="true">
        <intent-filter>
            <action android:name="android.intent.action.MAIN" />
            <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
    </activity>
```

MainActivity.java

```
public class MainActivity extends AppCompatActivity {

    private TextView display;
    private StringBuilder input;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.main_layout);

        display = findViewById(R.id.display);
        input = new StringBuilder();
    }

    public void onDigitClick(View view) {
        Button button = (Button) view;
        input.append(button.getText().toString());
        display.setText(input.toString());
    }

    public void onOperatorClick(View view) {
        Button button = (Button) view;
        input.append(" " + button.getText().toString() + " ");
        display.setText(input.toString());
    }

    public void onClearClick(View view) {
        input.setLength(0);
        display.setText("");
    }

    public void onEqualClick(View view) {
        try {
            double result = evaluateExpression(input.toString());
            display.setText(String.valueOf(result));
            input.setLength(0);
            input.append(result);
        } catch (Exception e) {
            display.setText("Error");
        }
    }

    private double evaluateExpression(String expression) {
        String[] tokens = expression.split(" ");
        if (tokens.length < 3) return 0;

        double num1 = Double.parseDouble(tokens[0]);
        String operator = tokens[1];
        double num2 = Double.parseDouble(tokens[2]);
```

```
        switch (operator) {
            case "+": return num1 + num2;
            case "-": return num1 - num2;
            case "*": return num1 * num2;
            case "/": return num2 != 0 ? num1 / num2 : 0;
            default: return 0;
        }
    }
```



8. Demonstrate use of scroll view.

AndroidMenifest.xml

```
<activity android:name=".MainActivity" android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>

</activity>

<activity android:name=".VeriticalScrlView" android:exported="true"/>
<activity android:name=".HorizontalScrlViewActivity" android:exported="true"/>
```

VeriticalScrlView.java

```
public class VeriticalScrlView extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.vertical_scrlview_layout);
    }
}
```

vertical_scrlview_layout.xml

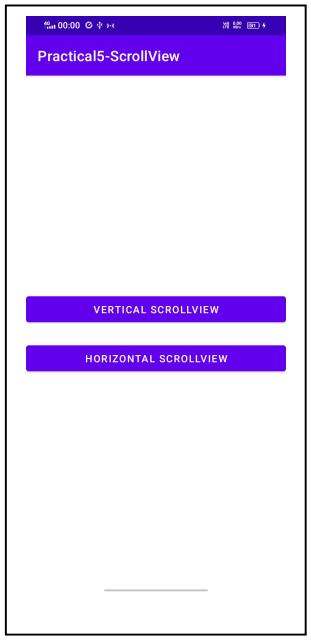
```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android=
"http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        >

        <TextView
            android:layout_width="match_parent"

android:layout_height="wrap_content"

android:background="@color/teal_700"
            android:gravity="center"
            android:text="Vertical ScrollView"
            android:textColor=
"@color/material_dynamic_neutral_variant100"
            android:textSize="30dp"
            android:textStyle="bold" />

        <TextView
            android:layout_width="match_parent"
            android:layout_height="335dp"
            android:layout_marginBottom="20dp"
            android:text="Hello World ...!!!"
            android:textSize="50dp"
            android:textStyle="bold" />

<TextView
android:layout_marginTop="70dp"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textStyle="bold"
android:textSize="20dp"
android:text="Sample of
VerticalScrollView"
android:layout_marginBottom="70dp" />

<TextView
android:layout_marginTop="70dp"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:textStyle="bold"
android:textSize="50dp"
android:text="Scroll Page"
android:layout_marginBottom="70dp" />

<TextView
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="70dp"
android:layout_marginBottom="70dp"
android:text="Hiii ... Enjoy android "
android:textSize="50dp"
android:textStyle="bold" />

</LinearLayout>

</ScrollView>
```

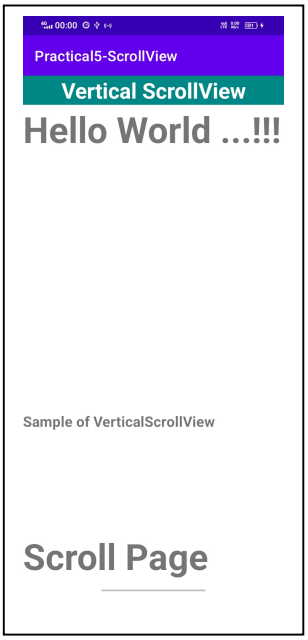
HorizontalScrlViewActivity.java

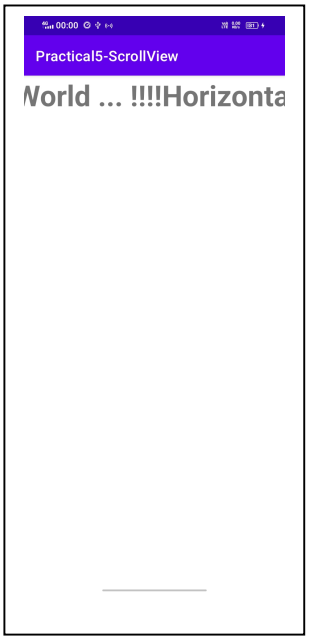
```java
public class HorizontalScrlViewActivity extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.horizontal_scrlview_layout);
    }
}
```

horizontal_scrlview_layout.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<HorizontalScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Hello World ... !!!!"
            android:textSize="40dp"
            android:textStyle="bold"
            />

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Horizontal ScrollView
"
            android:textSize="40dp"
            android:textStyle="bold"
            />
    </LinearLayout>
</HorizontalScrollView>
```







9. Demonstrate use of intent in android.

- **Explicit Intent**

AndroidMenifest.xml

```
<activity android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>
<activity android:name=".ShowDataActivity"/>
```

MainActivity.java

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);

        setContentView(R.layout.main_layout);
    }
    public void login(View v){

        Intent i = new Intent(MainActivity.this, ShowDataActivity.class);

        EditText username = findViewById(R.id.txtUsername);
        EditText password = findViewById(R.id.txtPassword);

        i.putExtra("keyUsername", username.getText().toString());
        i.putExtra("keyPassword",password.getText().toString());

        startActivity(i);
    }
}
```

main_layout.xml

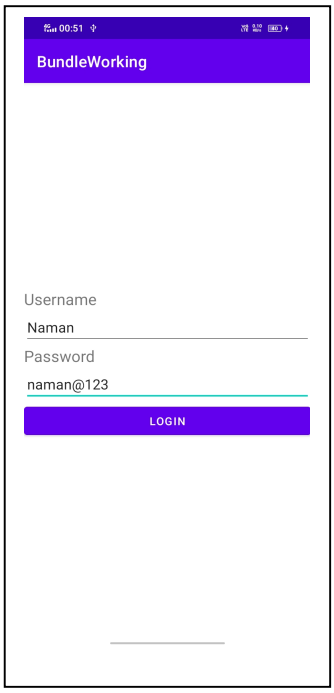
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android
="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center_vertical">

    <TextView
        android:id="@+id/lblUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Username"
        android:textSize="20sp" />

    <EditText
        android:id="@+id/txtUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        />

<TextView
android:id="@+id/lblPassword"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:text="Password"
android:textSize="20sp" />

<EditText
android:id="@+id/txtPassword"
android:layout width="match parent"
android:layout_height="wrap_content"
        />

<Button
android:layout width="match parent"
android:layout_height="wrap_content"
android:text="Login"
android:onClick="login"
 />

</LinearLayout>
```

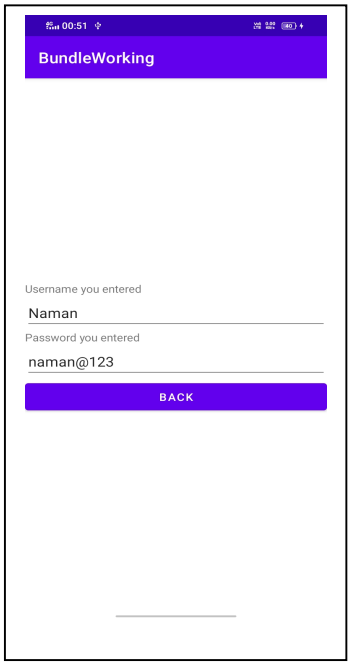
ShowDataActivity.java

```java
@Override
protected void onCreate(@Nullable Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);

    setContentView(R.layout.showdata_layout);

    EditText username = findViewById(R.id.txtSetUsername);
    EditText password = findViewById(R.id.txtSetPassword);

    Bundle bundle = getIntent().getExtras();

    username.setText(Objects.requireNonNull(bundle.get("keyUsername")).toString());
    password.setText(Objects.requireNonNull(bundle.get("keyPassword")).toString());

}
public void back(View v){
    startActivity(new Intent(ShowDataActivity.this, MainActivity.class));
}
```

showdata_layout.java

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android
="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center_vertical"
    >

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Username you entered "/>

    <EditText
        android:id="@+id/txtSetUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Password you entered”/>

    <EditText
        android:id="@+id/txtSetPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content" />

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Back"
        android:onClick="back" />
</LinearLayout>
```





- **Implicit Intent**

AndroidMenifest.xml

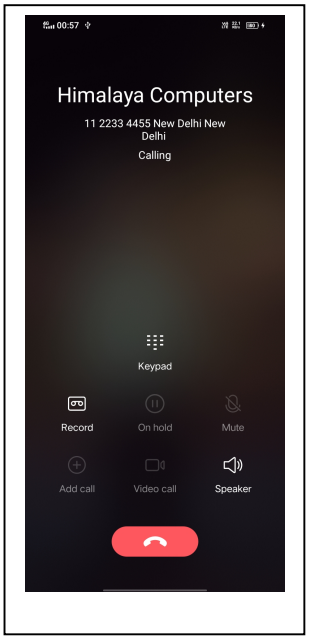
```
<uses-feature
    android:name="android.hardware.telephony"
    android:required="false" />

<uses-permission android:name="android.permission.CALL_PHONE"/>


<activity android:name=".MainActivity" android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>
```

MainActivity.java

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.main_layout);
    }
    public void call(View v){

        EditText txtMobileNo = findViewById(R.id.txtPhoneNumber);
        Intent i = new Intent(Intent.ACTION_CALL);
        i.setData(Uri.parse("tel:"+txtMobileNo.getText().toString()));
        startActivity(i);
    }
}
```

main_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android
="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center_vertical">

    <TextView
        android:id="@+id/lblphoneNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Phone Number "
        android:textSize="20sp" />

    <EditText
        android:id="@+id/txtPhoneNumber"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:inputType="phone" />

    <Button
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Call"
        android:onClick="call" />
</LinearLayout>
```



10. Create application to demonstrate menu option.

- **Option Menu**

AndroidMenifest.xml

```
<activity android:name=".MainActivity" android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>

<activity android:name=".OptionMenuActivity"/>
```

MainActivity.java

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.main_layout);
    }

    public void optionMenu(View v) {
       startActivity(new Intent(MainActivity.this,OptionMenuActivity.class));
   }
}
```

OptionMenuActivity.java

```
public class OptionMenuActivity extends AppCompatActivity {

    @Override
    protected void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
    }

    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.option_menu,menu);
        return super.onCreateOptionsMenu(menu);
    }

    @Override
    public boolean onOptionsItemSelected(@NonNull MenuItem item) {
        int id = item.getItemId();

        if(id == R.id.itemSetting){
            setContentView(R.layout.setting_layout);
            return true;
        }else{
            setContentView(R.layout.about_layout);
            return true;
        }
    }
}
```

main_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:orientation="vertical"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center" >

    <Button

        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Option Menu"
        android:onClick="optionMenu"
        />

</LinearLayout>
```

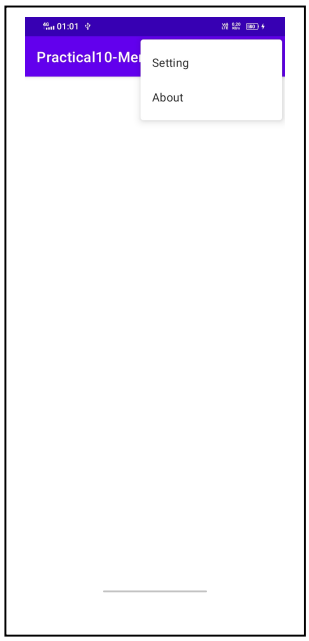
setting_layout.xml

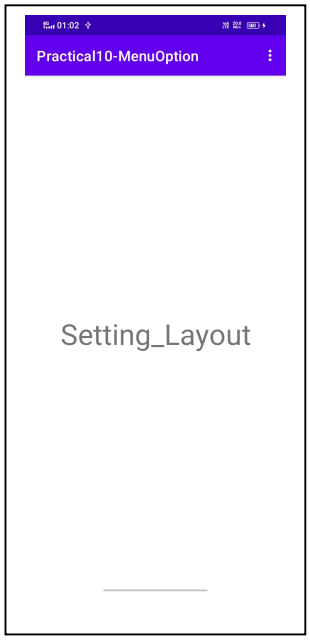
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    >

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="40dp"
        android:text="Setting_Layout"
        android:gravity="center"
        />

</LinearLayout>
```

about_layout.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    >

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="40dp"
        android:text="About_Layout"
        android:gravity="center"
        />

</LinearLayout>
```

OUTPUT :





- **Context Menu**

AndroidMenifest.xml

```
<activity android:name=".MainActivity"
    android:exported="true">
    <intent-filter>
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>
```

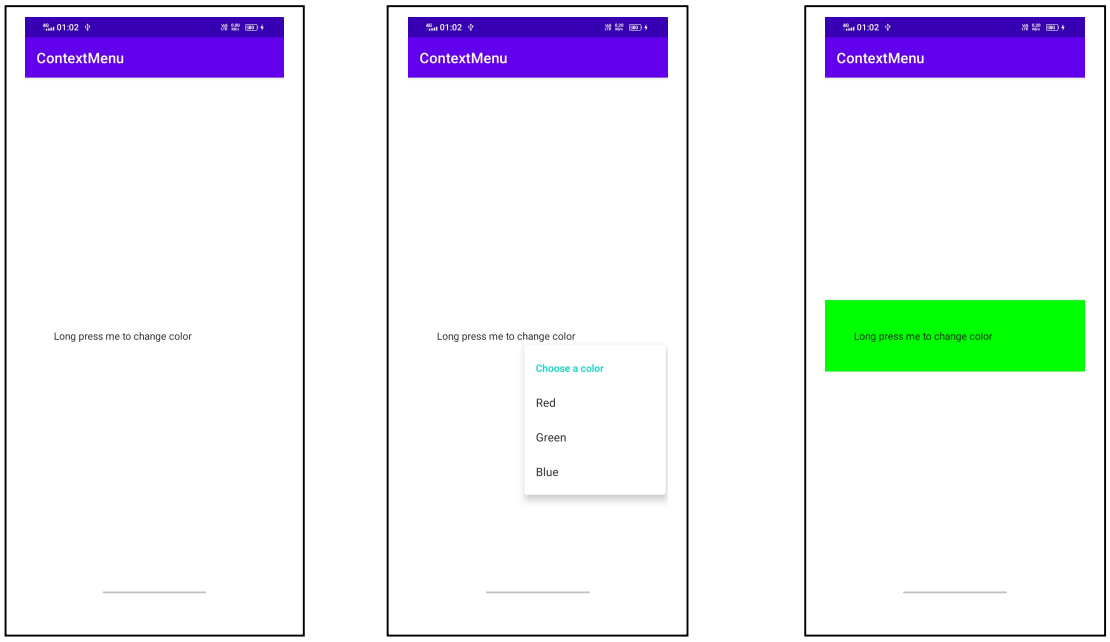
activity_main.xml

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    >

    <TextView
        android:id="@+id/textView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Long press me to change color"
        android:padding="40dp"
        style="@style/CardView.Dark"
        />
</LinearLayout>
```

MainActivity.java

```
public class MainActivity extends AppCompatActivity {

    private TextView textView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        textView = findViewById(R.id.textView);
        registerForContextMenu(textView); // Register the TextView for context
menu
    }

    @Override
    public void onCreateContextMenu(ContextMenu menu, View v,
ContextMenu.ContextMenuInfo menuInfo) {
        super.onCreateContextMenu(menu, v, menuInfo);
        menu.setHeaderTitle("Choose a color");
        menu.add(0, v.getId(), 0, "Red");
        menu.add(0, v.getId(), 0, "Green");
        menu.add(0, v.getId(), 0, "Blue");
    }

    @Override
    public boolean onContextItemSelected(MenuItem item) {
        switch (item.getTitle().toString()) {
            case "Red":
                textView.setBackgroundColor(Color.RED);
                break;
            case "Green":
                textView.setBackgroundColor(Color.GREEN);
                break;
            case "Blue":
                textView.setBackgroundColor(Color.BLUE);
                break;
        }
        return true;
    }
}
```

OUTPUT :



11. Create application to demonstrate progress bar.

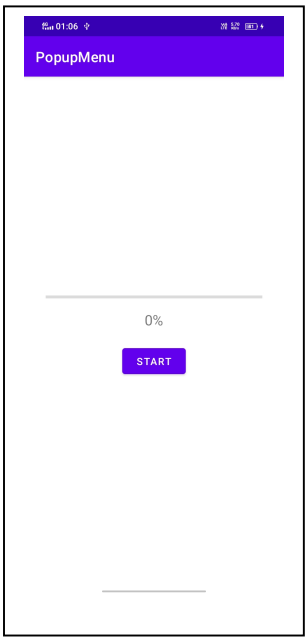
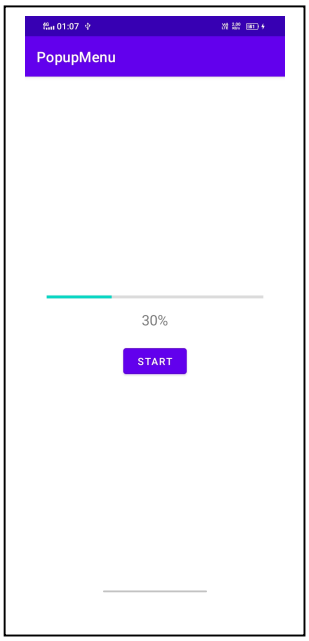
AndroidMenifest.xml

```
<activity android:name=".MainActivity" android:exported="true">
    <intent-filter >
        <action android:name="android.intent.action.MAIN"/>
        <category android:name="android.intent.category.LAUNCHER"/>
    </intent-filter>
</activity>
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    tools:context=".MainActivity">

    <ProgressBar
        android:id="@+id/progressBar"
        style="@style/Widget.AppCompat.ProgressBar.Horizontal"
        android:layout_width="300dp"
        android:layout_height="wrap_content"
        android:max="100"/>

    <TextView
        android:id="@+id/progressText"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="0%"
        android:textSize="20sp"
        android:layout_marginTop="10dp"/>

    <Button
        android:id="@+id/startButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

MainActivity.java

```java
package com.yogesh.popupmenu;
import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import android.widget.Button;
import android.widget.ProgressBar;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private ProgressBar progressBar;
    private TextView progressText;
    private int progressStatus = 0;
    private Handler handler = new Handler();

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        progressBar = findViewById(R.id.progressBar);
        progressText = findViewById(R.id.progressText);
        Button startButton = findViewById(R.id.startButton);

        startButton.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                startProgress();
            }
        });
    }

    private void startProgress() {
        progressStatus = 0;
        new Thread(new Runnable() {
            @Override
            public void run() {
                while (progressStatus < 100) {
                    progressStatus += 5;
                    handler.post(new Runnable() {
                        @Override
                        public void run() {
                            progressBar.setProgress(progressStatus);
                            progressText.setText(progressStatus + "%");
                        }
                    });
                    try {
                        Thread.sleep(300);
                    } catch (InterruptedException e) {
                        e.printStackTrace();
                    }
                }
            }
        }).start();
    }
}
```

OUTPUT :